

Cruciferous vegetables consumption and risk of renal cell carcinoma: a meta-analysis.

Previous cohort and case-control studies on the association between cruciferous vegetables consumption and risk of renal cell carcinoma have illustrated conflicting results so far. To demonstrate the potential association between them, a meta-analysis was performed. Eligible studies were retrieved via both computerized searches and review of references. The summary relative risks (RRs) with 95% confidence interval (CI) for the highest vs. the lowest consumption of cruciferous vegetables were calculated. Heterogeneity and publication bias were also evaluated. Stratified analyses were performed as well. Three cohort and 7 case-control studies were included. A significantly decreased risk with renal cell carcinoma was observed in overall cruciferous vegetables consumption group (RR = 0.73; 95% CI, 0.63-0.83) and subgroup of case-control studies (RR = 0.69; 95% CI, 0.60-0.78), but not in cohort studies (RR = 0.96; 95% CI, 0.71-1.21). No heterogeneity and publication bias were detected across studies. Our findings supported that cruciferous vegetables consumption was related to the decreased risk of renal cell carcinoma. Because of the limited number of studies, further well-designed prospective studies and researches need to be conducted to better clarify the protective effect of cruciferous vegetables on renal cell carcinoma and potential mechanism.